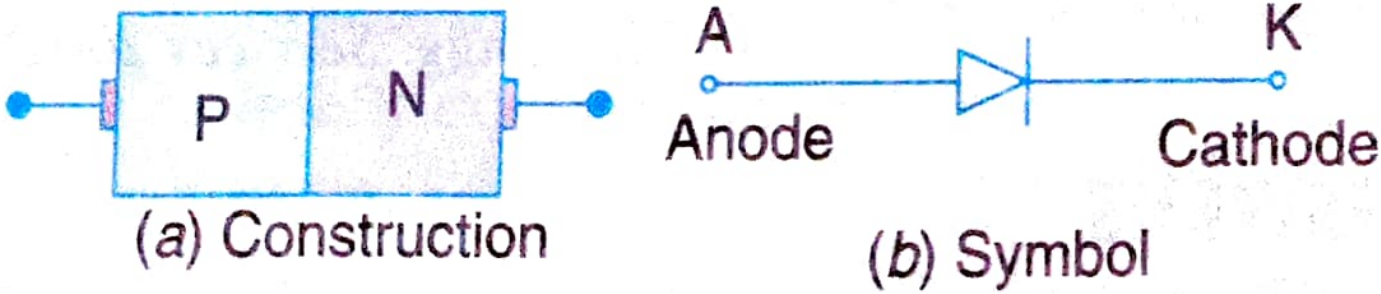
12.1. Introduction

A PN junction diode consists of a PN junction, formed either in *germanium* or *silicon* crystal. The diode has two terminals namely *anode* and *cathode*. The anode refers to the P-type region and cathode refers to the N-type region as shown in Fig. 12.1 (*a*). Its circuit symbol is as shown in Fig. 12.1 (*b*). The arrow head, shown in the circuit symbol, points the direction of current flow, when it is forward biased. It may be noted that it is the same direction in which the movement of holes takes place.



(*a*) Construction (*b*) Symbol

Fig. 12.1. PN junction diode.

12.2. V-I Characteristic of a PN Junction Diode

It is very important to know how a device responds (or behaves) when it is connected in an electrical circuit. This information is obtained by means of a graph known as its volt-ampere (or *V-I*) characteristics or simply characteristics.

It is a graph between the voltage applied across the terminals of a device and the current that flows through it. Figure 12.3 shows the *V-I* characteristics of a typical PN junction diode with respect to break down voltage (V_{BR}). It may be noted that the complete graph can be divided into two parts namely forward characteristic and reverse characteristic.

Now we shall discuss the forward characteristics and reverse characteristics of a PN junction diode along with the circuits for obtaining them in the laboratory.

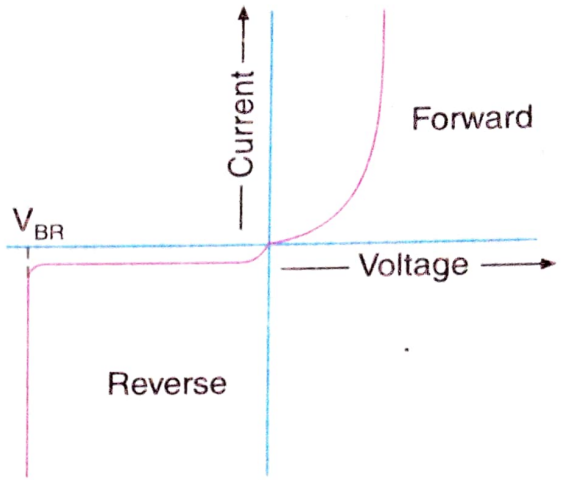
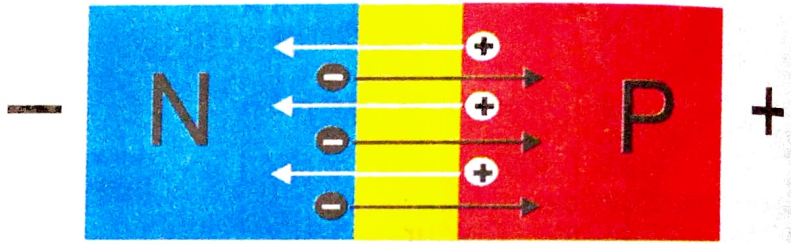


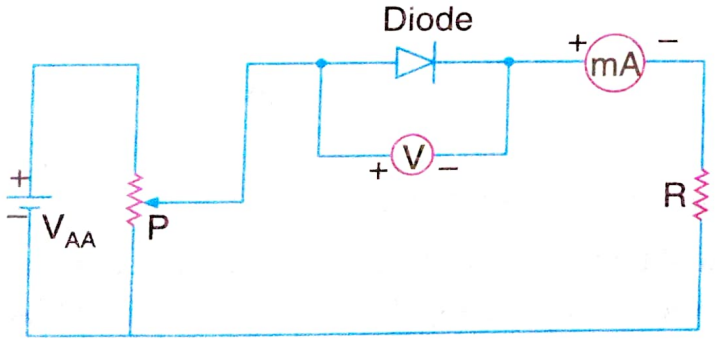
Fig. 12.3. VI characteristic of a *PN*-junction diode.

12.3. Forward Characteristic

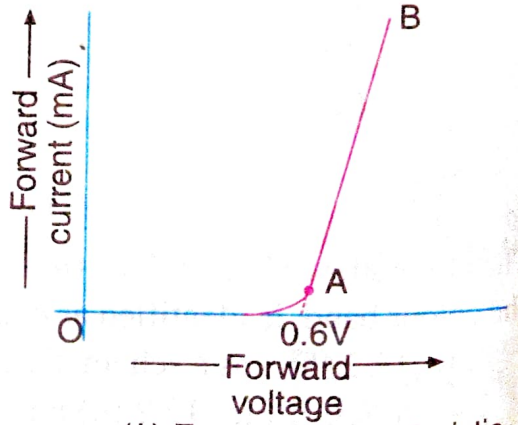
Figure 12.4 (*a*) shows the circuit arrangement for obtaining the forward characteristic of a diode. In this circuit, the diode is connected to a d.c. (V_{AA}) through a potentiometer (P) and a resistance (R). The potentiometer helps in varying the voltage applied across the diode. The resistance (R) is included in the circuit, so as to limit the current through the diode.



Forward biased diode.



(*a*) Circuit arrangement.



(*b*) Forward characteristic.

Fig. 12.4.

It will be interesting to know that if excessive current is permitted to flow through the diode, it may get permanently damaged. A voltmeter is connected across the diode to measure the voltage whereas a milliammeter measures current in the circuit.

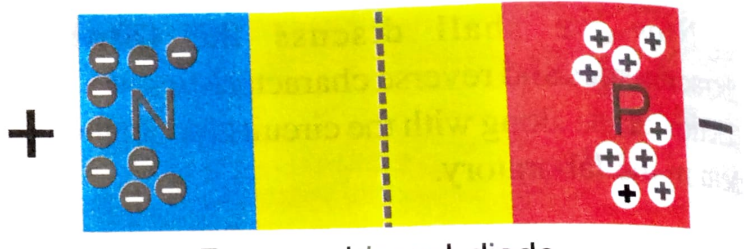
The positive terminal of the voltage source connected to the anode of a diode and negative terminal to the cathode. Hence the diode is forward-biased. Let us gradually increase the voltage in small steps of about 0.1 V and record the corresponding values of diode current. Now, if we plot a graph with voltage across the diode, along the horizontal axis and diode current, along the vertical axis, we shall obtain a curve *OAB* as shown in Fig. 12.4 (*b*). The curve *OAB* is called the forward characteristic of a silicon PN junction diode.

A careful study of the forward characteristic indicates that there is no diode current till the point *A* is reached. It is because of the fact, that the external applied voltage is being opposed by the junction voltage, whose value is 0.7 V for silicon and 0.3 V for germanium. However, as the voltage is increased above that of the point *A*, the diode current increases rapidly. It has been observed that a voltage of about 1 volt produces a forward current of about 20 to 50 mA. The applied voltage should not be increased beyond a certain safe limit, otherwise the diode is likely to burn out.

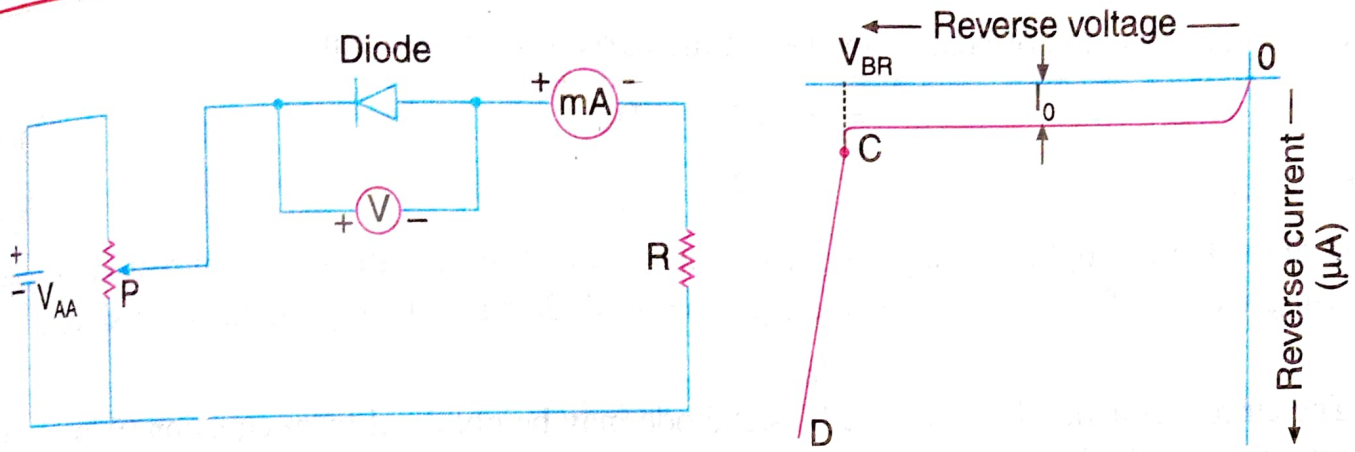
The voltage, at which the diode starts conducting, is called a knee voltage, cut-in voltage or threshold voltage. The knee voltage is designated either by V_K or V_g. Its value is equal to 0.6 V for silicon and 0.2 V for germanium. The knee voltage may be obtained from the forward characteristic by extending the curve *AB* backwards, till it meets the horizontal axis. The value on the horizontal axis is equal to the knee voltage.

12.4. Reverse Characteristic

The circuit arrangement for obtaining the reverse characteristic of a diode is as shown in Fig. 12.5 (*a*). This circuit is similar to that shown in Fig. 12.4, except two changes namely the diode terminals are reversed and the milliammeter is replaced by a microammeter.



Reverse biased diode.



(*a*) Circuit arrangement.

(*b*) Reverse characteristic.

Fig. 12.5.

It may be noted that negative terminal of the voltage source is connected to the anode of a diode and positive terminal to the cathode. Hence, the diode is reverse biased. The applied reverse voltage is gradually increased above zero in suitable steps and the values of diode current are recorded at each step. Now, if we plot a graph with reverse voltage along the horizontal axis and the diode current along the vertical axis, we shall obtain a curve marked *OCD* as shown in Fig. 12.5 (*b*). The curve *OCD* is called reverse characteristic of the diode.

A careful study of the reverse characteristic indicates that when the applied reverse voltage is below the breakdown voltage (V_{BR}), the diode current is small and remains constant. This value of current is called reverse saturation current (I_O). It is of the order of nanoamperes (1 $nA = 10^{-9}$A) for silicon diode and microamperes (1μA = 10^{-6} A) for germanium diode. When the reverse voltage is increased to a sufficiently large value, the diode reverse current increases as rapidly as shown by the curve *CD* in the figure. The applied reverse voltage, at which this happens, is known as break down voltage (V_{BR}) of a diode.

oltage (V_{BR}

12.5. Diode Current Equation

The mathematical equation, which describes the forward and reverse characteristic of a semiconductor diode is called the diode current equation.

Let I = Forward (or reverse) diode current,

I_0 = Reverse saturation current,

V = External voltage. It is positive for forward bias and negative for reverse bias.

η = A constant, whose value is equal to 1 for germanium diodes and 2 for silicon diodes for relatively low value of diode current (*i.e.*, at or below the knee of the curve) and $\eta = 1$ for germanium and silicon for higher levels of diode current (*i.e.*, in the rapidly increasing section of the curve)

V_T = Volt-equivalent of temperature. Its value is given by the relation, $\frac{T}{11\,600}$, where T is the absolute temperature. At room temperature (*i.e.*, 300 K), V_T is equal to 26 mV.

The current equation for a forward-biased diode is given by the relation,

$$I = I_0 \left(e^{V/\eta . V_T} - 1\right) \qquad \ldots (i)$$

We know that at room temperature, V_T is equal to 26 mV. Substituting the value of V_T (equal to 26 mV or 0.026 V) in equation (*i*),

$$I = I_0 \left(e^{40\,V/\eta} - 1\right)$$

$= 10.7\ \mu A$ **Ans.**

Example 12.2.

Determine the germanium PN junction diode current for the forward bias voltage of 0.22 V at room temperature 25°C with reverse saturation current equal to 1 mA. Take $\eta = 1$.

Solution.

Given : $V_F = 0.22$ volt; $T = 25°C = 298$ K; $I_0 = 1$ mA $= 1 \times 10^{-3}$ A and $\eta = 1$.

We know that volt equivalent of the temperature (T),

$$V_T = \frac{T}{11\,600} = \frac{298}{11\,600} = 25.7 \text{ mV} = 25.7 \times 10^{-3} \text{ V}$$

$\therefore$ The diode current,

$$I = I_0 \left(e^{V_F/\eta \cdot V_T} - 1\right)$$

$$= (1 \times 10^{-3}) \left(e^{0.22/(1 \times 25.7 \times 10^{-3})} - 1\right) \quad \ldots (\because \eta = 1 \text{ for germanium})$$

$$= (1 \times 10^{-3}) (e^{8.56} - 1) = 5.22 \text{ A}$$ **Ans.**

12.7. The Ideal Diode

We know that a PN junction diode conducts well, when it is forward biased and poorly (practically no current) when reverse biased. It will behave as an ideal, if the diode acts as a perfect conductor (with no voltage drop across it) when forward biased and as a perfect insulator (with no current through it) when reverse biased. The *V-I* characteristic of such an ideal diode is shown in Fig. 12.7 (*a*). An ideal diode acts like an automatic switch. The switch is closed when the diode is forward biased and is open when it is reverse biased as shown in Fig. 12.7 (*b*).

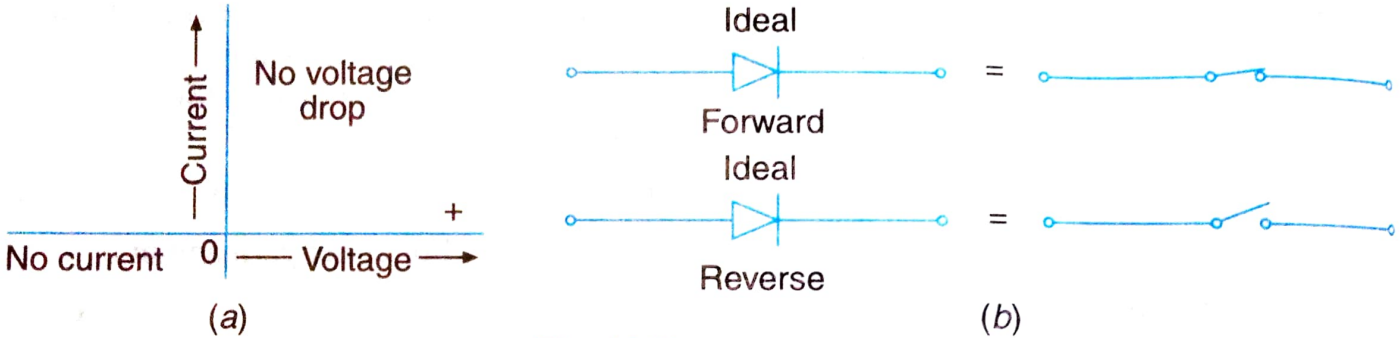


Fig. 12.7. Ideal diode.

In other words, an ideal diode is a device, which conducts with zero resistance when forward biased and appears as an infinite resistance when reverse biased. As a matter of fact, an ideal diode cannot be manufactured in actual practice. It is only a theoretical approximation of a real diode. However, in a well-designed electric circuit, a real diode behaves almost like an ideal diode because the forward voltage across the diode is small as compared to the input and output voltages.

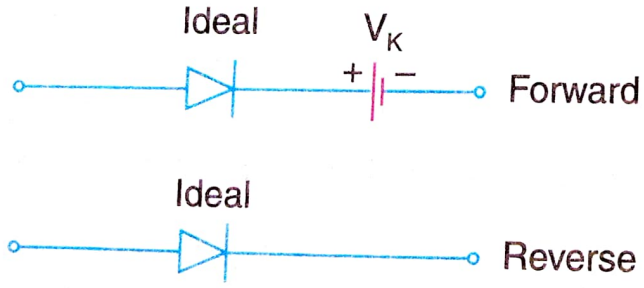


The ideal diode.

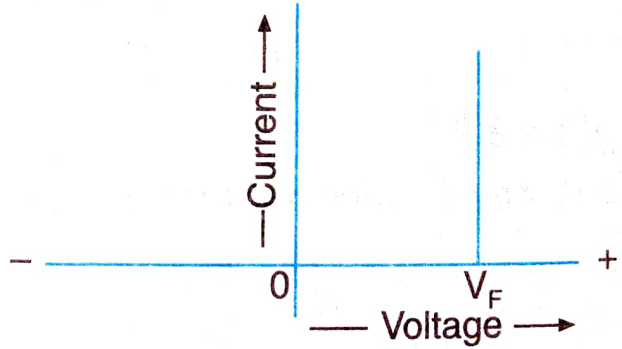
12.9. The Real Diode

We have already discussed in Art. 12.7 that a diode can be approximated to an automatic switch, which is closed when a diode is forward biased and open when it is reverse biased. But in actual practice, we have to consider two more factors, in addition to a simple switch, for a diode operation as discussed below:

When a diode is forward biased, it does not pass any current until the applied voltage exceeds the knee voltage or the cut-in voltage. Therefore, a diode may be represented by an automatic switch (or an ideal diode) in series with a battery of e.m.f. equal to the knee voltage. The knee voltage has no effect on the diode operation, when it is reverse biased. Therefore, the equivalent circuit of a real diode may be represented as shown in Fig. 12.9 (*a*) and its *V-I* characteristic as shown in Fig. 12.9 (*b*).



(*a*) Equivalent circuit.



(*b*) VI characteristic.

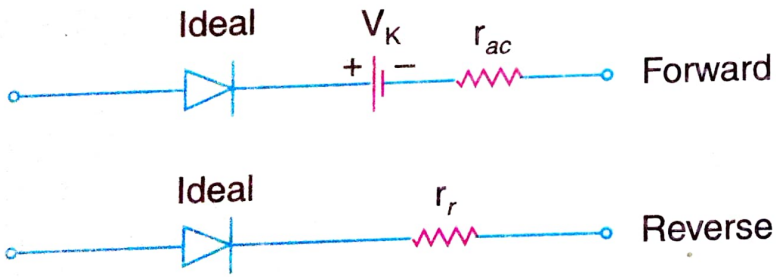
Fig. 12.9.

In addition to a knee voltage (V_K), a diode possesses a.c. or dynamic resistance, which is equal to the sum of bulk resistance and the junction resistance. Thus when a diode is forward biased, it does not conduct, until the applied voltage exceeds the knee voltage. As the applied voltage exceeds further, the current through a diode increases and produces a voltage drop across the a.c. resistance. Since the a.c. resistance is linear, therefore, the voltage drop across a real diode is given by the relation,

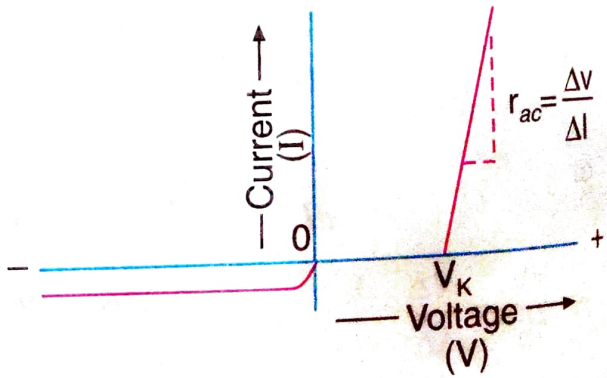
$$V_F = V_K + I_F \cdot r_{ac}$$

where
V_K = Knee voltage whose value is 0.6 V for silicon and 0.2 V for germanium,
I_F = Forward current, and
r_{ac} = A.C resistance.

When a diode is reverse biased, it does not behave like an open switch. In other words, the reverse current is not zero, though it has a small value. It is known as reverse saturation current. This means that a reverse biased diode is equivalent to an ideal diode in series with a high resistance (r_r) which is of the order of MΩ. Therefore, the equivalent circuit of a real diode may be represented as shown in Fig. 12.10 (*a*) and its *V-I* characteristic as shown in Fig. 12.10 (*b*).



(*a*) Equivalent circuit.



(*b*) V.I. characteristic.

Fig. 12.10. The real diode.

12.10. Diode Applications

A PN junction diode has an important characteristic that it conducts well in forward direction and poorly in reverse direction. This characteristic makes a diode very useful in a number of applications given below:

1. As rectifiers or power diodes in d.c. power supplies.
2. As signal diodes in communication circuits.
3. As zener diodes in voltage stabilizing circuits.
4. As varactor diodes in radio and TV receivers.
5. As a switch in logic circuits used in computers.

12.12. Diode Specification Sheets

All manufacturers of the semiconductor device provide data on specific diodes for the users to make proper utilization of the device. This data could be a brief description limited to a one page or more than that. It includes the information arranged in tables, graphs etc. The data is usually for:

1. Forward voltage, V_F (at a specific current and temperature).
2. Maximum forward current, I_F (at a specified temperature).
3. Reverse saturation current I_R or I_O (at a specified voltage and temperature).
4. Reverse voltage rating [*PIV*, *PRV* or *V*(*BR*)], where *BR* implies breakdown at a specified temperature.

5. Maximum power dissipation level at a particular temperature.
6. Capacitance values.
7. Reverse recovery time, t_{rr}.
8. Operating temperature range.

PN

MOTOROLA SEMICONDUCTORS

BOX 20912 PHOENIX ARIZONA 85036

Designers 'Data Sheat

"SURMETIC" ▲ RECTIFIERS

... subminiature size, axial lead mounted rectifiers for general- purpose low-power applications.

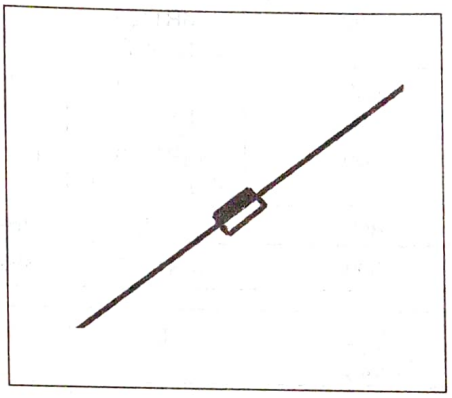
Designers Dat for "Wirst Case" Conditions

The Designers Data Sheets permit the design of most circuits entirely from the information presented. Limit curves–represented boundaries on device characteristics–are guveb ti facukutate "worst case" design.

1N4001 thru 1N4007

LEAD MOUNTED
LILICON RECTIFIERS

50-1000 VOLTS
DIFFUSED JUNCTION



*MAXIMUM RATING

Rating	Symbol	1N4001	1N4002	1N4003	1N4004	1N4005	1N4006	1N4007	Unit
Peak Repetitive Reverse Voltage Working Peak Reverse Voltage DC Blocking Voltage	V_{RRW} V_{RWM} V_R	50	100	200	400	600	800	1000	Volts
Non-Repetitive Peak Reverse Voltage (halfwave, single phase, 60 Hz)	V_{RWM}	60	120	240	480	720	1000	1200	Volts
RMS Reverse Voltage	$V_{R(RMS)}$	35	70	140	280	420	560	700	Volts
Average Rectified Forward Current (single phase, resistive load, 60 Hz, see Figure B, T_A = 75°C)	I_O	1.0							Amp
Non-Repetitive Peak Surge Current (surge applied at rated load conditions, see Figure 2)	I_{FMS}	30 (for 1 cycle)							Amp
Operating and Storage Junction Temperature Range	T_J . T_{stg}	– 65 to + 175							°C

*MAXIMUM RATING

Characteristic and Conditions	Symbol	Typ	Max	Unit
Maximum Instantaneous Forward Voltage Drop (I_F = 1.0 Amp, T_J = 25°C) Figure 1	V_F	0.93	1.1	Volts
Maximum Full-Cycle Average Forward Votage Drop I_O = 1.0 Amp, T_L = 75°C, 1 inch leads)	$V_{F(AV)}$	-	0.8	Volts
Maximum Reverse Current (rated dc voltage) T_J = 25°C T_J = 100°C	I_R	 0.05 1.0	 10 50	µA
Maximum Full-Cycle Average Reverse Current (I_O = 1.0 Amp, T_L = 75°C, 1 inch leads)	$I_{R(AV)}$	-	30	µA

*Indicates JEDEC Registered Data.

MECHANICAL CHARACTERISTICS

CASE: Void free, Transfer Molded

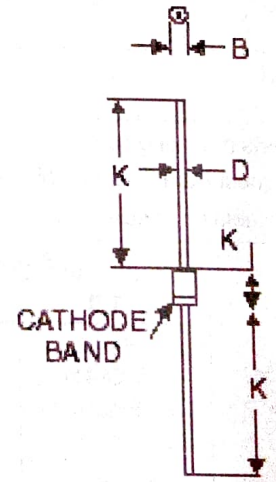
MAXIMUM LEAD TEMPERATURE FOR SOLDERING PURPOSES: 350°C
3/8" from case for 10 seconds at 5 lbs. tension

FINISH: All external surfaces are corrosion-resistant, leads are readily solderable

POLARITY: Cathode indicated by color band

WEIGHT: 0.40 Grams (approximately)

*Trademark of Motorola Inc.



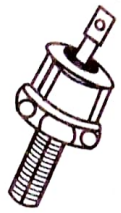
DIM	MILLIMETERS MIN	MILLIMETERS MAX	INCHES MIN	INCHES MAX
A	5.97	6.60	0.235	0.260
B	2.97	3.05	0.110	0.120
D	0.76	0.86	0.030	0.034
K	27.94	–	1.100	–

CASE 59-04

Doses Not Conform to Do-41 Outline

MOTOROLA INC. 1975 DS 6015 R3

Fig. 12.11 (*a*) Conrtesy of Motorola Inc.

VRRM (Volts)	I_O, AVERAGE RECTIFIED FORWARD CURRENT (Amperes)							
	12	20	24	25	30		40	50
	245A-02 (DO-203AA) Metal	Plastic Note 1	339-02 Plastic Note 2	193-04 (DO-02) Metal	43-02 (DO-203AB) Metal		42A-01 Metal	43-04
								
50	MR1120 1N1199, A, B	MR2000	MR2400	MR2500	1N3491	1N36659	1N1183A	MR5005
100	MR1121 1N1200, A, B	MR2001	MR2401	MR2501	1N3492	1N3660	1N1184A	MR5010
200	MR1122 1N1202, A, B	MR2002	MR2402	MR2502	1N3493	1N3661	1N1186A	MR5020
400	MR1124 1N1204, A, B	MR2004	MR2404	MR2504	1N3495	1N3663	1N1188A	MR5040
600	MR1126 1N1206, A, B	MR2006	MR2406	MR2506		Note 3	1N1190A	Note 3
800	MR1128	MR2008		MR2508		Note 3	Note 3	Note 3
1000	MR1130	MR2010		MR2510	Note 3	Note 3	Note 3	Note 3
I_{FMS} (Amps)	300	400	400	400	300	400	800	600
T_A @ Rated I_O (°C)								
T_C @ Rated I_O (°C)	150	150	125	150	130	100	150	150
T_J (Max) (°C)	190	175	175	175	175	175	190	195

Note 1. Meets mounting configuration of TO-220 outlline

Note 2. Request Data Sheet for Mounting Information.

Note 3. Available on special order.

VRRM (Volts)	I_O, AVERAGE RECTIFIED FORWARD CURRENT (Amperes)					
	1.0	1.5		3.0		6.0
	59-03 (DO-41) Plastic	59-04 Plastic	60-01 Metal	267-03 Plastic	267-02 Plastic	194-04 Plastic
						
50	†1N4001	**1N5391	1N4719	**MR500	1N5400	MR750
100	†1N4002	**1N5392	1N4720	**MR501	1N5401	MR751
200	†1N4003	1N5393 *MR5059	1N4721	**MR502	1N5402	MR752
400	†1N4004	1N5395 *MR5060	1N4722	**MR504	1N5404	MR754
600	†1N4005	1N5397 *MR5061	1N4723	**MR506	1N5406	MR756
800	†1N4006	1N5398	1N4724	MR508		MR758
1000	†1N4007	1N5399	1N4725	MR510		

(Continued)